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Roll No

ME-7004(1)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Machine Tool Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) List and describe commonly used attachments on lathe. 7
b) Describe the two methods of holding and driving straight shank twist drills. 7
2. Write down the important considerations in designs of speed regulation mechanism for machine tools. 14
3. a) Write the steps in design of power screw on the basis of strength, stiffness buckling. 10
b) Name power screw types and describe error compensation in power screws. 4
4. a) What are the steps involved in the proper lubrication? 8
b) Give the specification of lubrication oil. 6
5. a) What are sources of machine tool vibration? 5
b) Find an expression for static rigidity of a machine tool. 9

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6. Calculate the dimensions of all the gear parts of a single lead worm of 100mm outside diameter having pitch of 6mm. The ratio of worm to worm wheel is 80:1, face angle θ is 60° . 14
7. Why is it necessary to broach internal helical gear when the helix angle is greater than 15 degrees? 14
8. Write short notes :
- a) Bed and column sections in machine tools 3
 - b) Thick and thin film lubrication 4
 - c) Feed mechanism used in milling machine 4
 - d) Name the main elements of a lathe machine 3
